Proposal: Gmail Bulk Unsubscribe Assistant

Draft product requirements for a small tool that helps review and unsubscribe from mailing lists using Gmail message metadata, with browser fallback only when needed.

1. Goal

Build a simple tool that connects to a Gmail account, identifies mailing-list emails, groups them by sender/list, and lets the user safely unsubscribe in bulk with explicit review.

2. Problem

Gmail's built-in subscription management may not surface all subscriptions. Manually opening emails and finding unsubscribe links is slow, especially when hundreds of promotional messages come from recurring senders.

3. MVP Scope

Capability	Requirement
Connect to Gmail	OAuth-based Gmail access with least-privilege scopes.
Find subscriptions	Scan selected labels or recent mail and detect List-Unsubscribe headers plus unsubscribe links.
Group senders	Group by mailing list or sender and show approximate message count.
Review before action	User selects which subscriptions to remove. No automatic unsubscribe without confirmation.
One-click unsubscribe	Support standards-based HTTPS one-click unsubscribe when available.
Email unsubscribe	Support mailto-based unsubscribe when provided.
Browser fallback	Open a normal unsubscribe/preferences page when automated one-click is unavailable.
Protect important mail	Exclude financial, security, transactional, and user-protected senders from bulk actions by default.
Result tracking	Record success, failure, manual action required, and date attempted.

4. Suggested User Flow

1. Sign in with Google.

2. Choose a Gmail label or time range to scan.

3. Tool groups likely subscriptions by sender/list.

4. User reviews a table: sender, message count, unsubscribe method, and risk level.

5. User selects subscriptions and clicks Unsubscribe.

6. Tool performs standards-based unsubscribe actions where safe and opens browser pages for manual cases.

7. Tool shows a final status report.

5. Safety and Guardrails

• Never unsubscribe from a whole company/domain based only on domain name.

• Prefer mailing-list identity or the exact sender/list metadata.

• Require explicit user confirmation before executing unsubscribe actions.

• Do not automatically follow arbitrary links found in message bodies.

• Do not auto-unsubscribe financial institutions, security alerts, purchase confirmations, travel bookings, or other transactional mail unless the user explicitly selects that specific list.

• Store minimal metadata; avoid storing full email bodies unless needed.

6. Out of Scope for MVP

• General-purpose browser automation across every unsubscribe website.

• Automatic CAPTCHA handling.

• Automatic account-login flows on third-party sites.

• Deleting old emails.

• Creating Gmail filters or rules unless added as a separate feature later.

7. Recommended First Implementation

Backend: Python + Gmail API. **UI:** Streamlit or a small local web app. **Primary unsubscribe method:** parse List-Unsubscribe and List-Unsubscribe-Post headers. **Fallback:** open the unsubscribe/preferences URL in the user's browser.

8. Success Criteria

• User can review at least 50 recurring mailing lists in one screen.

• Standards-compliant one-click subscriptions can be unsubscribed without opening each email.

• Transactional/security senders are protected by default.

• Every action has a clear success/failure/manual status.

• The tool never unsubscribes a sender without explicit user selection.

Draft status: suitable for MVP planning. Detailed API design, OAuth scopes, data model, and implementation tasks can be added in the next revision.